

# Tuesday Reading Assessment: Chapters 3, 4-1 and 4-2

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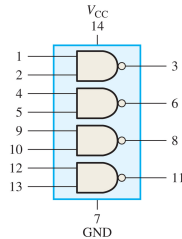


Figure 1: A 74xx00 gate, a quad NAND package IC.

## 1 Logic Gates and Logic Functions

- Figure 1 (left) contains a diagram of a quad-NAND IC package, the 74xx00 class. (a) Create and draw a system of NAND gates that is equivalent to a 2-input AND gate. (b) Determine a system of connections of outputs to inputs for the 74xx00 IC that will yield one AND gate. (c) Do the same for an OR gate.

## 2 Electrical Characteristics

- (a) What is the power dissipation of your AND gate made from NAND gates? Use  $P_D = V_{CC}(I_{CCH} + I_{CCL})/2$ ,  $I_{CCH} = 10 \mu A$ ,  $I_{CCL} = 100 \mu A$ , and  $V_{CC} = 5 V$ . (b) If the *propagation delay time*  $t_P$  is 5 ns per NAND gate, what is the **speed power product**,  $P_D t_P$ , for the AND gate in pJ? (c) What is the *fan-out* of your AND and OR gate designs from NANDs?

## 3 Programmable Logic: The AND Array

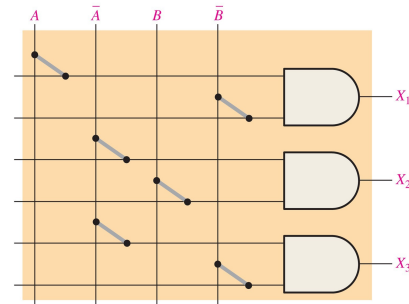


Figure 2: An example of the AND array within the broader category of PLAs.

- (a) What are the logical expressions for the AND array outputs in the PLA shown in Fig. 2? (b) Draw new connections such that:  $X_1 = \bar{A}B$ ,  $X_2 = AB$ , and  $X_3 = A\bar{B}$ .

## 4 Troubleshooting

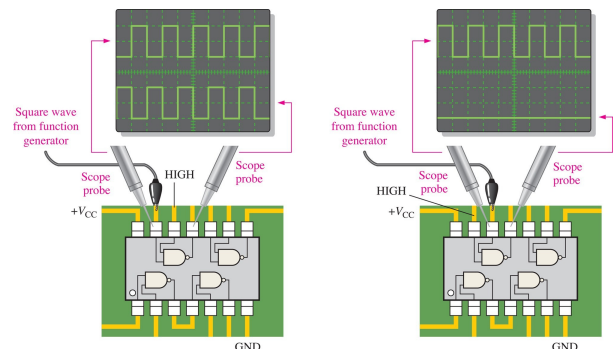


Figure 3: An example of probing a suspect gate with pulsed signal on one input, and HIGH on the other.

- Which input on the NAND gate pictured in the experiment in Fig. 3 is faulty?